(-1 for reversed arrows, -1 for extra level at 1.06)

3 (a) (i) constant amplitude B1

(ii) period = 0.75 s ...(allow ±0.2 s).. C1

$\omega = 2\pi/T$ C1

ω = 8.4 rad s^{-1} ...(-1 for 1 sf).. A1

(iii) either use of gradient or $v = \omega y_0$ C1

v = 0.168 m s^{-1} .... A1 [6]

(allow ±0.02 for construction: gradient drawn at wrong place 0/2)

(b) (i) 1.3 Hz B1

(ii) at ½f_0, ‘pulse’ provided to mass on alternate/some oscillations M1

so ‘pulses’ build up the amplitude .... A1 [3]